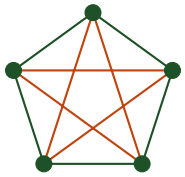
MID-SEMESTER STUDY SERIES GRAPH THEORY DAY 10

CS-5136: Graph Theory

Hamiltonian Graphs, Part 2 — Rebuilt, Chapter Complete

pages 189–194 · a degree-counting theorem (with its converse honestly checked), decomposing K_n, and a capstone homework



Built from: A. Bhattacharjee, "CS-5136: Graph Theory", NIT Silchar
this completes your mid-sem syllabus: pages 1–194, in full

1 A Degree-Counting Sufficient Condition

1.1 A warm-up: what does the condition even mean?

Before the formula, let's translate it into plain speech for small j. Read the theorem's hypothesis *"for every $1 \le j < n/2$, the number of vertices with degree $\le j$ is $< j$"* one value of j at a time:

- $j = 1$: the number of vertices with degree ≤ 1 must be < 1, i.e. **exactly zero**. In other words: no vertex is allowed to have degree 0 or 1.
- $j = 2$: the number of vertices with degree ≤ 2 must be < 2, i.e. **at most one** such vertex.
- $j = 3$: at most **two** vertices may have degree ≤ 3.

The pattern: low-degree vertices are allowed, but only in small, controlled numbers — you can't have *too many weak links* at once. This is a genuinely different flavour of condition from Ore's theorem (which looked at pairs); this one looks at how many low-degree vertices pile up overall.

Theorem 1.1

Let G have order $n \ge 3$. If, for every integer j with $1 \le j < \frac{n}{2}$, the number of vertices of G with degree at most j is *less than* j, then G is Hamiltonian.

Flagging this clearly: only one direction is claimed

Exactly like Ore's theorem, this is *not* an "if and only if" — it's a one-way sufficient condition. We'll prove that direction in full, then honestly test whether the reverse holds too (it won't — we already have the exact counterexample ready, reused from Day 9).

Proof. We use Day 9's closure theorem: G is Hamiltonian iff $C(G)$ is. Suppose, for contradiction, $C(G)$ is *not* complete (if it *were* complete, K_n is Hamiltonian for $n \ge 3$, and we'd be done immediately).

Setting up the extremal pair. Among all nonadjacent pairs in $C(G)$, let (u, v) maximize $d_{C(G)}(u) + d_{C(G)}(v)$. Since $C(G)$ is a closure (no more qualifying edges can be added), this maximum sum must be $\le n - 1$ — otherwise that very pair would have been joined already. Relabel so $d(u) \le d(v)$ (writing $d = d_{C(G)}$ from here on); then $2d(u) \le d(u) + d(v) \le n - 1$, so $d(u) \le \frac{n-1}{2} < \frac{n}{2}$. Write $k = d(u)$.

Why k is a legal choice of j. We need $1 \le k$. If the hypothesis holds, its $j = 1$ case forces $\delta(G) \ge 2$ (no vertex of degree ≤ 1) — and closure only raises degrees, so $\delta(C(G)) \ge 2$ too, giving $k = d(u) \ge 2 \ge 1$. Combined with $k < n/2$ from above, k is exactly the kind of j the hypothesis talks about.

The key claim: every vertex nonadjacent to v has degree $\le k$. Suppose some x nonadjacent to v had $d(x) > k = d(u)$. Then (x, v) is itself a nonadjacent pair with $d(x) + d(v) > d(u) + d(v)$ — contradicting that (u, v) was chosen to *maximize* this sum. So no such x exists.

Counting. There are exactly $(n - 1) - d(v)$ vertices other than v that are nonadjacent to it, and *every one of them* has degree $\le k$ by the claim just proved (this includes u itself). So:

$$\#\{\text{vertices of degree} \le k\} \;\ge\; (n - 1) - d(v).$$

Since $d(u) + d(v) \le n - 1$ and $d(u) = k$, we get $d(v) \le n - 1 - k$, so

$$(n - 1) - d(v) \;\ge\; (n - 1) - (n - 1 - k) \;=\; k.$$

Combining: $\#\{\text{vertices of degree} \le k\} \ge k$.

Contradiction. But the hypothesis (at $j = k$, which we showed is legal) says this count must be **strictly less than** k. We've just shown it's $\ge k$. Contradiction. So $C(G)$ must be complete after all, hence Hamiltonian, hence (Day 9) G is Hamiltonian. ∎

Example — Checking the converse honestly: does it fail?

Take $G = C_5$ again (the same graph that broke Ore's converse in Day 9 — worth noticing it's a repeat offender). C_5 is obviously Hamiltonian. Check the hypothesis: $n = 5$, so we need $j = 1, 2$ (since $j < 2.5$). $j = 1$: no vertex has degree ≤ 1 (all have degree 2) — count is $0 < 1$ ✓. $j = 2$: **every** vertex has degree ≤ 2 — count is 5. Is $5 < 2$? **No.** The hypothesis **fails** at $j = 2$, even though C_5 is Hamiltonian. So once again: a Hamiltonian graph exists that fails the sufficient condition completely — the converse is false here too, for exactly the same underlying reason as Ore's theorem (both conditions are generous shortcuts, not universal requirements).

2 Counting Hamiltonian Cycles in K_n

2.1 A warm-up: how many Hamiltonian cycles could possibly fit?

Before building anything, let's just count. K_5 has $\binom{5}{2} = 10$ edges. A single Hamiltonian cycle on 5 vertices uses exactly 5 edges. So at most $10/5 = 2$ completely edge-disjoint Hamiltonian cycles could possibly exist side by side (using more would need more edges than K_5 even has). This simple counting argument is exactly half of the theorem below — the other half is showing 2 is actually *achievable*, not just an upper limit.

Theorem 2.1

For odd $n \geq 3$, K_n decomposes into $\frac{n-1}{2}$ edge-disjoint Hamiltonian cycles.

This is a construction/existence statement, not an equivalence, so there's no separate "converse direction" to prove — but it does split naturally into two halves (upper bound, then a matching construction), and we'll do both fully.

Proof (Half 1: the upper bound, generalizing the warm-up). K_n has $\binom{n}{2} = \frac{n(n-1)}{2}$ edges, and every Hamiltonian cycle uses exactly n of them. So the number of *edge-disjoint* Hamiltonian cycles that could possibly fit is at most

$$\frac{n(n-1)/2}{n} = \frac{n-1}{2}.$$

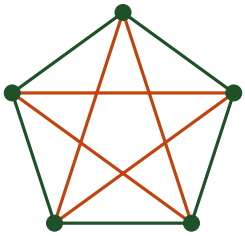
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Proof (Half 2: a construction meeting the bound exactly). Fix one vertex v_0, and place the remaining $m = n - 1$ vertices (an *even* number, since n is odd) labelled $1, 2, \ldots, m$ evenly around a circle. Build a Hamiltonian cycle by zigzagging outward from the two ends:

$$v_0,\ 1,\ m,\ 2,\ m{-}1,\ 3,\ m{-}2,\ \ldots,\ v_0$$

(alternately taking the smallest and largest remaining label). This uses each "chord length" between consecutive labels exactly once. Now **rotate** the labels $1, \ldots, m$ by $\frac{360^\circ}{m}$, $2 \cdot \frac{360^\circ}{m}$, …, $\frac{n-3}{2} \cdot \frac{360^\circ}{m}$ (keeping v_0 fixed): each rotation relabels the same zigzag pattern onto a genuinely different set of chords, producing a new Hamiltonian cycle sharing no edge with any earlier one. Together with the original, this gives $\frac{n-3}{2} + 1 = \frac{n-1}{2}$ edge-disjoint Hamiltonian cycles — meeting the upper bound exactly, so this decomposes *all* of K_n's edges. ■

Example — Fully verified: K_5 into 2 Hamiltonian cycles



$n = 5$: predicted $\frac{5-1}{2} = 2$ cycles, matching our warm-up's upper bound exactly. **Green (the pentagon)**: $0, 1, 2, 3, 4, 0$ — the 5 "short" edges. **Rust (the pentagram)**: $0, 2, 4, 1, 3, 0$ — the 5 "long" edges. Checked directly: these two edge sets are disjoint, and together they are *exactly* all $\binom{5}{2} = 10$ edges of K_5. ✓ (The same zigzag-and-rotate construction was also verified computationally for K_7 and K_9.)

3 A Capstone Homework

Homework

Let G be 3-regular of order 12, and H be 4-regular of order 11. Is $G + H$ (the join, Day 8) Eulerian? Is it Hamiltonian?

Example — Both questions resolved using tools from this whole series

Eulerian? Using Day 8's join formula: $G+H$ is Eulerian iff every degree in G matches $|V(H)| = 11$'s parity, and every degree in H matches $|V(G)| = 12$'s parity. G is 3-regular (3 is odd) and 11 is odd — match. H is 4-regular (4 is even) and 12 is even — match. **Yes, $G + H$ is Eulerian**, and concretely: every G-vertex has degree $3 + 11 = 14$ in the join, every H-vertex has degree $4 + 12 = 16$ — both even, and the join is always connected.

Hamiltonian? Order $n = 12 + 11 = 23$. In a join, every G-vertex is adjacent to every H-vertex — so the *only* possible nonadjacent pairs are within G alone or within H alone. For a nonadjacent pair inside G: degree sum $= (3 + 11) + (3 + 11) = 28 \geq 23 = n$. For a nonadjacent pair inside H: degree sum $= (4 + 12) + (4 + 12) = 32 \geq 23 = n$. Every nonadjacent pair meets Ore's threshold (Day 9) — **yes, $G + H$ is Hamiltonian** too, by Ore's Theorem directly. (And note: this doesn't need the converse of Ore's theorem, which we know is false — we're using the *forward* direction, exactly as proved, which is all that's ever needed to *conclude* Hamiltonicity.)

4 Day 10 Cheat-Sheet

Recap: which theorems are "iff" and which aren't

- **One direction only (converse checked and found false — C_5 again):** the degree-counting theorem. Same underlying reason as Ore's theorem in Day 9: sufficient conditions are generous shortcuts, not strict requirements.
- **Existence/construction, no converse to ask:** K_n (n odd) decomposes into exactly $\frac{n-1}{2}$ edge-disjoint Hamiltonian cycles — fix one vertex, zigzag the rest, rotate.
- $j = 1$'s case alone forces $\delta(G) \geq 2$ whenever the degree-counting hypothesis holds.
- Join $G+H$: connected always; Eulerian iff degree-parities cross-match the other side's order (Day 8); often Hamiltonian too, since only within-G or within-H pairs can possibly be nonadjacent, making Ore's theorem easy to satisfy.

5 Practice Questions

Set A — the degree-counting theorem

Q1. A graph has order 10. For the hypothesis to possibly apply, which values of j need checking?

A. $1 \leq j < 5$, *i.e.* $j = 1, 2, 3, 4$.

Q2. A graph of order 9 has exactly one vertex of degree 2 and no vertex of degree ≤ 1; every other vertex has degree ≥ 5. Does the theorem apply?

A. *Check* $j = 1, 2, 3, 4$ *(since* $n/2 = 4.5$*).* $j = 1$*: 0 vertices of degree* ≤ 1*, need* < 1 ✓*.* $j = 2$*: 1 vertex of degree* ≤ 2*, need* < 2 ✓*.* $j = 3, 4$*: same single vertex still the only one* ≤ 3 *or* ≤ 4*, so* $1 < 3$ *and* $1 < 4$ ✓*. All pass — Hamiltonian.*

Set B — decomposing K_n

Q3. How many edge-disjoint Hamiltonian cycles decompose K_7? How many edges does each use, and does it total correctly?

A. $\frac{7-1}{2} = 3$ *cycles, each using* 7 *edges:* $3 \times 7 = 21 = \binom{7}{2}$*, matching* K_7*'s total edge count exactly.*

Q4. Why does Theorem 2.1 require n to be *odd*?

A. *If* n *is even, every vertex of* K_n *has* odd *degree* $n-1$ *— but a Hamiltonian cycle gives every vertex degree exactly* 2 *within it, so decomposing all of* K_n*'s edges into whole Hamiltonian cycles alone can't work out (you'd need to cover an odd leftover degree using only even contributions). Even-*n *complete graphs need a different decomposition (Hamiltonian cycles plus one leftover perfect matching) — outside this course's scope.*

Set C — the capstone, and one last honest check

Q5. In the capstone homework, would the join still be Hamiltonian if G were only 2-regular (instead of 3-regular), everything else the same?

A. *Recompute: nonadjacent pair inside* G *now has degree sum* $(2+11)+(2+11) = 26$*, still* $\geq n = 23$ *— yes, still Hamiltonian by Ore's theorem. There's real slack in this construction.*

Q6. Across Days 9 and 10, name every theorem we found to be *one-directional with a false converse*, and the single graph that broke every single one of them.

A. *Ore's Theorem (Day 9) and the degree-counting theorem (today) — both broken by the exact same graph, C_5: Hamiltonian, yet failing each theorem's hypothesis completely. A nice reminder that "easy to verify, but not universally necessary" is a common and useful shape for a sufficient condition to take.*

The syllabus — complete

That closes pages 1–194 — everything from the Shortest Superstring Problem on Day 1 to this capstone join example today, across ten sessions: graphs and subgraphs, degrees and degree sequences, walks and connectivity, Euler tours, and Hamiltonian cycles, with every sketchy proof completed, every homework resolved, every "if and only if" proved in both directions, and every one-directional theorem's converse honestly checked (and, where false, disproved with a real counterexample) along the way. Good luck with the mid-semester exam.